

A Few Series

1. Does $\sum_{k=0}^{\infty} \frac{2^k}{3^k}$ converge? If so, what is the limit?

2. Does $\left\{ \frac{2^k}{3^k} \right\}$ converge? If so, what is the limit?

Do the following series converge or diverge? State the test you used. For convergent series, indicate if they converge conditionally or absolutely. If appropriate, find the sum.

3. $\sum_{k=1}^{\infty} \frac{k-1}{k2^k}$

4. $\sum_{k=1}^{\infty} \frac{7^k}{k^2}$

5. $\sum_{k=1}^{\infty} (-1)^k \frac{k^2}{k!}$

6. $\sum_{k=1}^{\infty} k e^{-k}$

7. $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$

8. $\sum_{k=1}^{\infty} \frac{\sin k}{k\sqrt{k}}$

9. $\sum_{k=2}^{\infty} \frac{1 + \ln k}{k^3}$

10. $\sum_{k=1}^{\infty} \frac{\cos k\pi}{k+3}$

11. $\sum_{k=2}^{\infty} \frac{3^{k-1} 2^{k+2}}{7^{k+1}}$

12. $\sum_{k=1}^{\infty} \frac{\ln k}{\sqrt{3k+2}}$

13. Find all values of x for which the series $\sum_{k=3}^{\infty} \frac{k(x-5)^k}{3^k}$ converges.

Yet More Series

1. Do the following series converge or diverge? State the test you used. For convergent series with negative terms, indicate if they converge conditionally or absolutely. If appropriate, find the sum.

(a) $\sum_{k=1}^{\infty} \left(\frac{k+4}{3k-1} \right)^k$

(b) $\sum_{k=2}^{\infty} \frac{k^3 - 3k}{1 + \ln k}$

(c) $\sum_{k=3}^{\infty} \frac{1}{k} - \frac{1}{k+1}$

(d) $\sum_{k=1}^{\infty} k \frac{3^k}{4^k}$

(e) $\sum_{k=0}^{\infty} \frac{(-1)^k}{2\sqrt{k} + 3}$

(f) $\sum_{k=2}^{\infty} \frac{k!}{k2^k}$

(g) $\sum_{k=2}^{\infty} \frac{3}{k \ln k}$

(h) $\sum_{k=2}^{\infty} \frac{3(-1)^k}{k \ln k}$

2. What is the difference between absolute and conditional convergence?

3. What is the difference between a sequence and a series?

4. Suppose $a_k > 0$ and $\lim_{k \rightarrow \infty} \frac{a_k}{1/k} = \frac{1}{2}$. Does $\sum_{k=0}^{\infty} a_k$ converge or diverge? Why?

5. Suppose $a_k > 0$ and $\frac{a_{k+1}}{a_k} = \frac{3}{k}$. Does $\sum_{k=1}^{\infty} a_k$ converge or diverge? Why?